

Refurbish a Returned Unit SOP (1x Episode: 1 Unit)

One episode refurbishes one returned unit. The unit is slid out of its used poly bag, wiped with a damp cloth, given a new stock label laid over its old one, slid into a new poly bag and sealed, and set in the stock bin. Its two accessories are put back in the accessory tray before the unit is restocked. The table begins with the **returns tote** in its **start zone**, holding the bagged unit, a **coiled cable**, and a **manual card**. The **cloth tray** sits on the **right side** at mid depth and the **label pad** at the **front-right corner**. The **bag stack** sits at the **front-left** and the **scrap bin** at the **back-left**. The **accessory tray** sits at the **front-center-right** and the **stock bin** at the **front-center**. The center of the table is the **working area**: its left half is the **bag spot** and its right half is the **work spot**.

The order never changes: unbag, wipe, label, bag, reset the accessories, restock. No label is taken before the wipe is finished. No bag is laid out before the new label is pressed. The accessories are not touched until the bag is sealed, and the unit is not restocked until both accessories are in the tray.

Once the unit is out of its used bag it **stays flat on the table**. It is slid, never lifted and never turned over, until it is lifted inside its sealed new bag in Step 6.

The right gripper does the work: it slides the unit out of the used bag, wipes it, presses the new label, slides it into the new bag, seals the flap, sets both accessories in their wells, and lifts the bagged unit into the stock bin. The left gripper holds: it holds the used bag flat, bins it, holds the unit at its left end through the wipe and the label, lays out the new bag and holds its mouth open, holds the bag body for the seal, and holds the stock bin down for the set-down. The left gripper works the left side, the center, the front-center, and (Config L and M) the start zone. The right gripper works the right side, the center, the front-center-right, and (Config R) the start zone. Which gripper takes things out of the returns tote depends on the config.

The table is set up in one of three ways. Only the **returns tote** moves; the cloth tray, the label pad, the bag stack, the scrap bin, the accessory tray, the stock bin, and the working area are in the same place in all three.

- **Config L:** the returns tote is at the **left side, mid depth**, between the scrap bin and the bag stack.
- **Config M:** the returns tote is at the **back-center**, behind the working area.
- **Config R:** the returns tote is at the **back-right**.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the returns tote starts at the left side, mid depth (**Config L**), the back-center (**Config M**), or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the tote's side takes everything out of it — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Hand-over rule:** in Config L and M the accessory tray is too far right for the left gripper, so it **hands each accessory over** to the right gripper above the work spot: the left gripper holds the accessory still, the right gripper closes on the opposite side of it, and only then does the left gripper open. The right gripper lowers it into its well. Nothing is handed over in Config R. The bagged unit is never handed over: the gripper that lifts it out of the tote lays it on the bag spot.
- **Fixed roles:** everything else is the same in all three — the right gripper slides, wipes, labels, bags, seals, and restocks; the left gripper holds the bag, the unit, the mouth, and the bin; and the used bag, the new bag, the cloth, the label, the accessories, and the sealed bag go to the same places.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the returns tote in its start zone, the cloth tray on the right side, the label pad at the front-right corner, the bag stack at the front-left, the scrap bin at the back-left, the accessory tray at the front-center-right, the stock bin at the front-center, and the whole working area at the center.
3. The working area is visible from above, so the **label panel** on the unit's top face, every wipe pass, the **bag mouth**, and the **bag flap** can all be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the returns tote, the cloth tray, the label pad, the bag stack, the scrap bin, the accessory tray, the stock bin, and robot hardware.
6. The right arm reaches the cloth tray, the label pad, the accessory tray, the inside of the stock bin, and (Config R) the inside of the returns tote without stretching. The left arm reaches the bag stack, the scrap bin, the left end of the unit on the work spot, the bag spot, the left rim of the stock bin, and (Config L and M) the inside of the returns tote without stretching.

Materials checklist

1. One open-topped **returns tote** sits in the **start zone** for this episode's config, **square** to the front edge, and the other two start zones are bare. It holds one returned **unit** inside a used **poly bag**, one **coiled cable**, and one **manual card**, all lying loose and not stacked on each other.
 - **Config L:** left side, mid depth, between the scrap bin and the bag stack
 - **Config M:** back-center, behind the working area
 - **Config R:** back-right
2. The **unit** is a flat boxy device about the size of a hardback book. It lies in the tote with its long side left to right and its **label panel** facing up. One closed gripper can pinch it across its short side.
3. The unit's top face carries a printed **label panel**, centered, and one **old label** is stuck inside that panel.
4. The used poly bag arrives **open**: its **bag flap** is already peeled back and its **bag mouth** stands clear, so no flap is peeled during an episode.
5. The **coiled cable** is already coiled and held by its **tie**, with the tie facing up.
6. The **manual card** is a stiff printed card lying **printed side up**.
7. The **cloth tray** sits on the **right side** at mid depth and holds one damp **wipe cloth**, folded flat, with its near edge free of the tray floor so it can be pinched.
8. The **label pad** sits at the **front-right corner** and holds a liner of printed **stock labels**, printed face up. The top label's near edge is already lifted free of the liner. A stock label is bigger than the old label on every side, so one laid square over the old label covers it.
9. The **bag stack** sits at the **front-left** and holds new **clear poly bags** lying flat, all facing the same way, each with its **bag flap** lying back flat and its **adhesive strip** exposed, clean, and facing up.
10. The **scrap bin** sits at the **back-left**, open on top and empty.
11. The **accessory tray** sits at the **front-center-right**, square to the front edge. It is shallow and has two printed wells: the **cable well** at the back and the **card well** at the front. Both are empty.
12. The **stock bin** sits at the **front-center**, open on top, square to the front edge, and holds **two** finished bagged units lying flat, labels facing up.
13. Before collection, confirm by hand that:
 - the unit slides out of the used bag mouth without the bag tearing or lifting;
 - a new bag's mouth stands open when its top wall is lifted at the left of the mouth;
 - the unit slides all the way into a new bag with room to spare past the mouth;
 - a stock label comes away from the liner on a straight lift and covers the old label with room on every side;

- a folded flap sticks down across its full width and stays down when the bag body is nudged;
- the coiled cable sits inside the cable well and the manual card sits flat inside the card well, neither riding a rim;
- a third bagged unit lies flat in the stock bin without riding the bin rim.

Workspace layout

- **Working area:** the center of the table — every slide, wipe, label press, insert, and seal happens here
- **Bag spot:** the left half of the working area — the used bag is emptied here and the new bag is laid out, filled, and sealed here
- **Work spot:** the right half of the working area — the unit sits here bare while it is wiped and labeled
- **Start zone:** the returns tote — left side, mid depth (**Config L**), back-center (**Config M**), or back-right (**Config R**)
- **Right side, mid depth:** the cloth tray
- **Front-right corner:** the label pad
- **Front-left:** the bag stack
- **Back-left:** the scrap bin
- **Front-center-right:** the accessory tray
- **Front-center:** the stock bin

Arm assignments

- **Left gripper:** holds the used bag flat while the unit comes out, drops the used bag in the scrap bin, holds the unit at its left end through the wipe and the label, lays out the new bag and holds its mouth open, holds the bag body through the seal, and holds the stock bin down by its left rim. In Config L and M it also brings the bagged unit in from the tote, and lifts each accessory out of the tote and hands it to the right gripper above the work spot.
- **Right gripper:** slides the unit out of the used bag, wipes it, takes and presses the new stock label, slides the unit into the new bag, seals the flap, sets the cable and the manual card in the accessory tray, and lifts the sealed bag into the stock bin. In Config R it also brings the bagged unit in from the tote and lifts each accessory out of the tote; in Config L and M it receives each accessory from the left gripper above the work spot.

Vocabulary

- **Working area:** the center of the table where the unit is handled. It has two halves, the **bag spot** on the left and the **work spot** on the right.

- **Bag spot:** the left half of the working area. The used bag is emptied here and the new bag is laid out here.
- **Work spot:** the right half of the working area. The bare unit sits here for the wipe and the label.
- **Start zone:** where the returns tote sits at the start of the episode — left side, mid depth (**Config L**), back-center (**Config M**), or back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Square:** the long side of the unit, tote, tray, or bin lines up with the front edge of the table, so neither end sits nearer the front.
- **Unit:** the returned device being refurbished. One unit per episode.
- **Label panel:** the printed outline on the unit's top face that a label goes inside.
- **Old label:** the label already stuck inside the label panel when the unit arrives. It is never peeled off — the new label goes over it.
- **Stock label:** one printed adhesive label from the label pad. One goes on the unit each episode.
- **Lifted edge:** the near edge of the top stock label on the pad, already free of the liner. It is the only part of a label a gripper ever touches.
- **Covered:** no printing from the old label shows at any edge of the new stock label.
- **Seated:** the stock label lies inside the label panel, printing facing up, flat with no lifted corner and no bubble, and stays down when the right gripper comes away.
- **Press:** the closed right gripper set down flat on the label or the flap and held still for 2 seconds.
- **Wipe cloth:** the damp folded cloth in the cloth tray. It touches the unit only, and it goes back in the tray when the wipe is done.
- **Pass:** one straight drag of the cloth across a face of the unit, held down on the face the whole way.
- **Poly bag:** one flat clear bag holding one unit. Its **bag mouth** is the open end, its **top wall** is the upper face of the bag beside the mouth, and its **bag flap** is the lip beyond the mouth, carrying an exposed adhesive strip on its upper face. The strip is exposed from staging, so no flap is ever peeled during an episode.
- **Proud of the mouth:** part of the unit still sticks out past the bag mouth, so the flap cannot fold over a clear mouth.
- **Sealed bag:** the flap folded over the bag mouth onto the top wall and pressed along its full width, with the adhesive strip taken hold, no corner lifting, and the flap staying down when the bag body is nudged.
- **Flat hold:** the closed left gripper laid down on a bag and held there, pinning it to the table without pinching it.

- **Low carry:** the item is carried just clear of the table, tray rims, or bin rims, never lifted high above them.
- **Coiled cable:** the cable that came back with the unit, already coiled and held by its **tie**.
- **Tie:** the strap around the coiled cable. It is what the right gripper pinches, and it faces up when the cable is in its well.
- **Manual card:** the stiff printed card that came back with the unit. It stays printed side up the whole episode.
- **Hand over:** the left gripper holds an accessory still above the work spot, the right gripper closes on the opposite side of it, and only then does the left gripper open and lift clear. Config L and M only, because the accessory tray is too far right for the left gripper.
- **Accessory tray:** the shallow two-well tray at the front-center-right. The **cable well** is the back well and the **card well** is the front well.
- **In its well:** the accessory lies inside the printed well, flat on the well floor, with no part of it riding the well rim or lying outside the tray.
- **Stock bin:** the open-topped bin at the front-center that finished bagged units are restocked into.
- **Restocked:** the sealed bag lies flat in the stock bin on top of the units already there, its stock label facing up and readable through the bag, square to the bin walls, with no part riding the bin rim.

Steps

Run Steps 1–6 in order on the one unit, then end the episode with Step 7. Steps 1.1, 5.1, and 5.2 depend on where the returns tote is: in Config L and M the **left gripper** takes the bagged unit and each accessory out of the tote and **hands each accessory over** to the right gripper above the work spot; in Config R the **right gripper** takes them out itself. Every other line is the same in all three configs.

Step 1: Unbag the returned unit

Goal: the unit lies bare and flat on the work spot, and the used poly bag is in the scrap bin.

- Keep every carry **low** over the table and over the tote rim.
- Once the unit is out of the bag it stays flat on the table. Do not lift it and do not turn it over.

1.1 Bring the bagged unit in

Look where the returns tote is before reaching for the unit.

- **IF the returns tote is at the left side (Config L):** with the **left gripper**, close on the bagged unit over the unit inside it, so the jaws hold the unit through the bag. Lift it straight up, just clear of the tote rim, and carry it low and right into the working area.

- **IF the returns tote is at the back-center (Config M):** with the **left gripper**, close on the bagged unit over the unit inside it, so the jaws hold the unit through the bag. Lift it straight up, just clear of the tote rim, and carry it low and forward into the working area.
- **IF the returns tote is at the back-right (Config R):** with the **right gripper**, close on the bagged unit over the unit inside it, so the jaws hold the unit through the bag. Lift it straight up, just clear of the tote rim, and carry it low and left into the working area.

Then, in all three:

- Lay it flat with the **bag body on the bag spot** and the **bag mouth facing right**, toward the work spot, with the unit's **label panel facing up**.
- Open the gripper and come away.

1.2 Slide the unit out

- With the **left gripper**, take a **flat hold** on the bag body left of the **bag mouth** and hold it down on the bag spot.
- With the **right gripper**, reach in through the **bag mouth**, close on the **right end** of the unit, and slide it straight right out of the mouth onto the **work spot**.
- Keep the unit flat on the table the whole way. Do not lift it out.
- Open the **right gripper** and come away up and to the right.

1.3 Bin the used bag

- With the **left gripper**, let go of the flat hold, close on the used bag, and lift it off the bag spot.
- Carry it back and left to the **scrap bin** and drop it in.
- Bring the **left gripper** back to the working area.

Check: the unit lies flat and bare on the work spot, square to the front edge, its **label panel** facing up with the **old label** showing; the bag spot is bare; the used bag is in the scrap bin. If the unit does not slide, lay the bag flatter with the **left gripper** and slide again. If the bag comes off the bag spot with the unit, lay the bag back flat with the **left gripper** and slide the unit the rest of the way out.

Expected state: the unit is on the work spot, the bag spot is bare, the scrap bin holds one used bag, and the returns tote still holds the coiled cable and the manual card.

Step 2: Wipe the unit

Goal: the unit's top face and front face have each been wiped with the damp cloth.

- With the **left gripper**, close on the **left end** of the unit, jaws on its front and back faces, and hold it down against the table. Keep the **left gripper** there through Steps 2 and 3.
- With the **right gripper**, pinch the **wipe cloth** by its near edge in the **cloth tray**, lift it straight up clear of the tray rim, and carry it low to the work spot.

2.1 Wipe the top face

- Lay the cloth flat on the unit's top face at its **back edge**, at the **left third** of the face.
- Draw it straight toward the front edge in one **pass**, keeping it pressed on the face the whole way, until it runs off the front edge of the face.
- Lift the cloth, set it back down at the back edge one step to the right, and make the next pass the same way.
- Make **three passes** in all — left third, middle, right third — each overlapping the one before, so the whole top face is covered.

2.2 Wipe the front face

- Hold the cloth against the unit's **front face** and draw it once from **left to right**, keeping it pressed on the face the whole way.

2.3 Put the cloth back

- With the **right gripper**, carry the cloth low back to the **cloth tray** and lay it flat inside it, then open the gripper and come away.

Check: three passes ran back to front across the top face, stepping left to right and overlapping each other, one pass ran left to right down the front face, and the cloth lies flat in the cloth tray. If the cloth lifts off a face part-way through a pass, lay it down again at the start of that pass and make that pass again. If the unit slides under the cloth, hold it again with the **left gripper** and repeat the pass.

Expected state: the unit is still on the work spot, held at its left end, label panel facing up with only the old label on it; the cloth is flat in its tray.

Step 3: Apply the new label over the old

Goal: one stock label is **seated** in the label panel and the old label is **covered**.

- With the **left gripper**, keep holding the unit at its left end.
- With the **right gripper**, close on the **lifted edge** of the top label on the **label pad** and lift straight up, so the label comes away from the liner.
- Carry the label in one go to the unit's top face, sticky face down. Do not set it down on the way and do not let it touch the table.
- Lower it straight down over the **old label**, printing facing up, its edges square with the unit's front edge and inside the **label panel**, then open the **right gripper**.
- Close the **right gripper** and set it down flat on the label. **Press** for 2 seconds, then lift straight up.

Check: the label is **seated** and the old label is **covered** — no old printing shows at any edge, the new label is flat with no lifted corner and no bubble, and it stays down when the right gripper comes away. If a corner lifts, press that corner again with the closed **right gripper** for 2

seconds. If any old printing still shows at an edge, take a second stock label and lay it over the first, lined up to cover what shows, then press it for 2 seconds. Never peel the old label off and never peel back a label that is already down.

Expected state: the unit carries one seated stock label with no old printing showing, the unit is still flat on the work spot, and the label pad still lies flat at the front-right corner.

Step 4: Bag the unit in a new poly bag

Goal: the unit lies inside a new clear poly bag with the flap sealed and the stock label readable through the bag.

4.1 Lay the new bag out

- With the **left gripper**, close on the near edge of the top bag on the **bag stack** at the front-left and lift it clear of the stack.
- Lay it flat on the **bag spot**, long side left to right, with its **bag mouth** and **bag flap** facing **right**, toward the work spot, and its flap lying back flat with the **adhesive strip** facing up.
- Take one bag per trip. The **right gripper** waits on its own side while the bag comes in.

4.2 Slide the unit in

- With the **left gripper**, pinch the **top wall** just left of the **bag mouth** and lift it just enough for the mouth to stand open.
- With the **right gripper**, close on the **right end** of the unit and slide it straight **left** into the open mouth, keeping it flat on the table the whole way.
- Slide until the whole unit is inside and clear of the mouth, then open the **right gripper** inside the bag and come away up and to the right.
- Do not lift the unit and do not turn it. It goes in flat, label panel up, the same way it lay on the work spot.

4.3 Seal the flap

- With the **left gripper**, take a **flat hold** on the bag body left of the mouth and keep it for the whole seal.
- With the **right gripper**, close on the outer edge of the **bag flap**, fold it **left** over the **bag mouth**, and lay it down on the **top wall** so the adhesive strip meets the bag.
- Press along the flap from its back end to its front end until the strip has taken hold across the full width and no corner is lifting.
- The flap is folded once and laid down once. Do not peel a stuck flap back to lay it again — the strip loses its hold.

Check: the whole unit is inside with no part **proud of the mouth**, the bag lies flat on the bag spot, the flap lies flat over the mouth with the strip taken hold across its full width and no corner lifting, the flap stays down when the bag body is nudged, and the stock label reads through the

top wall. If the unit is proud of the mouth, hold the mouth open again with the **left gripper** and slide it the rest of the way in with the **right gripper** before folding the flap. If a corner of the flap lifts, press that corner again with the closed **right gripper** for 2 seconds.

Expected state: one **sealed bag** lies flat on the bag spot with the unit inside, label facing up; the work spot is bare; the bag stack still lies flat at the front-left.

Step 5: Reset the accessories

Goal: the coiled cable and the manual card each lie **in its well** in the accessory tray, and the returns tote is empty.

- With the **left gripper**, let go of the bag and hold clear of the front-center.
- Move **one accessory at a time** and keep every carry **low**. Never drop an accessory into a well from above and never tip the tote out.

5.1 Reset the cable

- **IF Config L or M:** with the **left gripper**, pinch the **coiled cable** in the returns tote by its **tie**, lift it straight up clear of the tote rim, carry it low to above the **work spot**, and hold it still; the **right gripper** closes on the coil on the side opposite the tie, then the left gripper opens and lifts clear. **IF Config R:** with the **right gripper**, pinch the **coiled cable** in the returns tote by its **tie** and lift it straight up, clear of the tote rim.
- With the **right gripper**, carry it low to the **accessory tray** and lower it into the **cable well** with the **tie facing up**, then open the gripper and come away.

5.2 Reset the manual card

- **IF Config L or M:** with the **left gripper**, pinch the **manual card** in the returns tote by its near edge, lift it straight up clear of the tote rim, carry it low and flat to above the **work spot**, and hold it still, **printed side up**; the **right gripper** closes on the opposite edge, then the left gripper opens and lifts clear. **IF Config R:** with the **right gripper**, pinch the **manual card** in the returns tote by its near edge and lift it straight up, clear of the tote rim.
- With the **right gripper**, carry it low to the **accessory tray**, keeping it flat and **printed side up**, and lay it flat in the **card well**, square to the well, then open the gripper and come away.

Check: the cable lies in the **cable well** with its tie facing up, the card lies flat in the **card well** printed side up and square to it, neither rides a well rim or lies outside the tray, and the returns tote is empty. If an accessory lands on a rim or outside its well, lift it with the **right gripper** and set it down again in its well. If anything is still in the tote, lift it out and set it in its well.

Expected state: the accessory tray holds the cable in the back well and the card in the front well, the returns tote is empty, and the sealed bag still lies flat on the bag spot.

Step 6: Restock the bin

Goal: the sealed bag is **restocked** in the stock bin.

- With the **left gripper**, press down on the **left rim** of the **stock bin** at the front-center and hold it there.
- With the **right gripper**, close on the bag over the **right end of the unit** inside it, so the jaws hold the unit through the bag.
- Lift it straight up, just clear of the table, and keep it level. Do not turn it in the air.
- Carry it forward to the **stock bin** and lower it in until it rests on the unit already there, **label facing up**, square to the bin walls, lying the same way round as it lay on the bag spot.
- Open the **right gripper** and come away.
- Keep the **left gripper** on the bin rim until the right gripper is clear.

Check: the bag lies flat in the stock bin on top of the two units already there, its stock label facing up and readable through the bag, square to the bin walls, with no part riding the bin rim. If it rides the rim or lands crooked, pinch it again with the **right gripper**, lift it clear, and lower it again.

Expected state: the stock bin holds three bagged units lying flat, labels facing up; the working area is bare; the returns tote is empty; the accessory tray holds both accessories.

Step 7: End the episode

Goal: recording ends with the unit wiped, relabeled, rebagged, restocked, and both accessories reset.

1. Confirm the end state:
 - the stock bin holds three bagged units lying flat, the newest on top, every label facing up;
 - the new unit's stock label is **seated** and the old label is **covered**, readable through the bag;
 - that unit's bag is a **sealed bag** with the flap flat across its full width;
 - the working area is bare and nothing is loose on the table;
 - the used poly bag is in the scrap bin and the returns tote is empty;
 - the coiled cable is in the cable well tie up and the manual card is flat in the card well printed side up;
 - the wipe cloth lies flat in the cloth tray and the label pad, bag stack, and trays are still on their spots.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Lift the newest bagged unit out of the stock bin. Peel its flap back, slide the unit out, and peel the stock label off the label panel. Throw the used bag and the peeled label away.
2. Leave the **old label** on the unit. If the old label came away with the stock label, stick a fresh old label inside the label panel.
3. Take the used bag out of the scrap bin and throw it away, so the scrap bin is empty again.
4. Take the coiled cable and the manual card out of the accessory tray. Leave both wells empty.
5. Put the unit back in the **returns tote**: slide it into a used poly bag with its flap peeled back and its mouth open, label panel facing up, and lay it in the tote square to the front edge. Lay the coiled cable beside it with its tie up, and the manual card beside it printed side up, all loose and not stacked. Set the tote in the start zone for the next episode's config — left side, mid depth (Config L), back-center (Config M), or back-right (Config R) — square to the front edge, leaving the other two start zones bare.
6. Check the stock bin holds two bagged units again, lying flat, labels facing up.
7. Lay the wipe cloth flat in the cloth tray, near edge free. Damp it again if it has dried out.
8. Lift the near edge of the new top label on the label pad free of the liner.
9. Square the bag stack at the front-left, every bag facing the same way, each flap lying back flat with its adhesive strip facing up.
10. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the unit, the accessories, and the fixtures. Replace the unit if its case is cracked, its label panel is worn, or a label no longer sticks to it. Replace the cable if its tie no longer holds the coil, and the manual card if it is creased, curled, or no longer readable. Replace a poly bag whose adhesive strip has been touched or has lost its tack, and throw out any bag that is torn.
2. Wash or replace the wipe cloth. Fit a fresh liner on the label pad if it is running low. Top the bag stack back up.
3. Leave the returns tote holding one bagged unit, one coiled cable, and one manual card in the start zone for the next session's first config; the cloth tray holding one damp folded cloth; a fresh label lifted on the label pad; the bag stack squared at the front-left; the scrap bin empty at the back-left; both wells of the accessory tray empty at the front-center-right; and the stock bin holding two bagged units at the front-center.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R (returns tote at the back-right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the returns tote is not in the start zone for the config; a gripper reaches across the table into the tote; in Config L or M the left gripper carries an accessory to the accessory tray itself, or the right gripper reaches into the tote for one, instead of a hand-over above the work spot; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the same-side rule, and the hand-over rule (the gripper on the tote's side takes the bagged unit and each accessory out — the left gripper in Config L and M, the right gripper in Config R; in Config L and M each accessory is handed over to the right gripper above the work spot; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the returns tote is before the first reach, then follow that config's IF lines through Steps 1.1, 5.1, and 5.2.

Violation: Unit worked off the work spot

- **Visible cue:** the right gripper wipes the unit or presses the stock label while the unit is still in the returns tote, still inside its used bag, on the bag spot, or anywhere else on the table rather than on the work spot.
- **SOP rule broken:** Steps 1.2, 2, and 3 (the unit is slid out onto the work spot and is wiped and labeled there).

- **Coaching note:** get the unit out and onto the work spot first. Nothing is wiped or labeled through a bag or over a tote.

Violation: Used poly bag not binned, or reused

- **Visible cue:** the episode ends with the used bag on the table, on the bag spot, back in the returns tote, or in the stock bin; or the same bag the unit came out of is used again to bag it.
- **SOP rule broken:** Step 1.3 (the left gripper drops the used bag in the scrap bin) and Step 4.1 (the new bag comes off the bag stack).
- **Coaching note:** the old bag goes straight in the scrap bin. Every unit goes out in a fresh bag.

Violation: Unit lifted off the table or turned over

- **Visible cue:** between coming out of the used bag and being lifted in its sealed new bag, the right gripper lifts the unit clear of the table, tips it up on an edge, or turns it so the label panel is not facing up.
- **SOP rule broken:** Steps 1.2, 2, 3, and 4.2 (once the unit is out of its used bag it stays flat on the table and is slid, not lifted; it is not lifted again until Step 6).
- **Coaching note:** slide it, never carry it. The label panel faces up from the moment it leaves the bag.

Violation: Unit not held during the wipe or the label

- **Visible cue:** the left gripper is off the unit's left end while the right gripper wipes or presses the label, and the unit slides, spins, or is chased across the work spot.
- **SOP rule broken:** Step 2 (the left gripper closes on the unit's left end and holds it there through Steps 2 and 3).
- **Coaching note:** left gripper on the left end first, and leave it there until the label is pressed.

Violation: Unit not fully wiped

- **Visible cue:** fewer than three passes run across the top face, the passes leave a strip of the face untouched between them, a pass starts short of the back edge or stops short of the front edge, the cloth lifts off the face part-way through a pass, or the front face is never wiped.
- **SOP rule broken:** Steps 2.1 and 2.2 (three overlapping passes back to front across the top face, stepping left to right, then one pass left to right down the front face, cloth held on the face the whole way).
- **Coaching note:** three passes across the top, overlap each one, then one down the front. Keep the cloth pressed on the face all the way through.

Violation: Cloth mishandled

- **Visible cue:** the right gripper wipes with a bare gripper, wipes with the cloth over the label pad, the accessory tray, or the table, drops the cloth and carries on wiping with it, or leaves the cloth on the unit, on the table, or in the working area at the end.
- **SOP rule broken:** Steps 2 and 2.3 (the cloth comes out of the cloth tray, touches the unit only, and goes back flat in the cloth tray).
- **Coaching note:** cloth on the unit, then straight back in the tray. If you drop it, stop and reset — do not wipe with it again.

Violation: Old label not covered

- **Visible cue:** at the end of Step 3, or through the bag at the end of the episode, printing from the old label still shows at an edge of the new stock label.
- **SOP rule broken:** Step 3 (lower the stock label square over the old label so no old printing shows, and lay a second label over what shows if any does).
- **Coaching note:** line the new label up over the old one before you let go. If any old printing peeks out, put a second label over it.

Violation: Stock label not seated

- **Visible cue:** the label sits outside the label panel, crooked to the unit's front edge, upside down, or with a lifted corner or a bubble, or the right gripper never presses it flat for a full 2 seconds.
- **SOP rule broken:** Step 3 (lower the label into the label panel printing up and square to the front edge, then press flat for 2 seconds; press a lifted corner again).
- **Coaching note:** square it up before you let go, then hold the press for a full two seconds.

Violation: Label mishandled on the way

- **Visible cue:** the right gripper takes more than one label off the liner, touches the sticky face, folds or creases the label, sets it down on the way, or lets it touch the table.
- **SOP rule broken:** Step 3 (close on the lifted edge only, and carry the label in one go to the unit, sticky face down).
- **Coaching note:** lifted edge only, one label, straight to the panel.

Violation: A label peeled off

- **Visible cue:** either gripper peels the old label off the label panel, or peels a stock label back off after it is down and lays it again.
- **SOP rule broken:** Step 3 (the old label is never peeled off, and a label that is already down is never peeled back).
- **Coaching note:** the new label goes over the old one. Once a label is down it stays down — cover it, do not lift it.

Violation: New bag laid out wrong

- **Visible cue:** the bag goes down on the bag spot with its mouth and flap facing away from the work spot, folded under the bag, or with the adhesive strip face down; or more than one bag comes off the bag stack in a trip.
- **SOP rule broken:** Step 4.1 (lay one bag flat on the bag spot, mouth and flap facing right toward the work spot, flap lying back flat with its strip facing up).
- **Coaching note:** one bag, mouth toward the unit, flap back flat and strip up before anything goes in.

Violation: Bag mouth not held open

- **Visible cue:** the left gripper is off the top wall while the right gripper slides the unit in, the unit is shoved against a closed mouth, or the bag is dragged across the bag spot by the unit.
- **SOP rule broken:** Step 4.2 (the left gripper pinches the top wall left of the mouth and lifts it so the mouth stands open while the unit goes in).
- **Coaching note:** open the mouth first, hold it open, then slide. The bag does not move, the unit does.

Violation: Unit not fully in the bag

- **Visible cue:** part of the unit is **proud of the mouth** when the right gripper comes away, or the flap is folded over with the unit still sticking out of the mouth.
- **SOP rule broken:** Step 4.2 (slide until the whole unit is inside and clear of the mouth before the flap is folded).
- **Coaching note:** all the way in and clear of the mouth, then seal. Check the mouth before you fold.

Violation: Flap not sealed, or the seal reworked

- **Visible cue:** the episode moves on with the flap still lying back open, the strip taken hold on only part of the width, or a corner lifting; or the right gripper peels a stuck flap back off and lays it down a second time.
- **SOP rule broken:** Step 4.3 (fold the flap once over the mouth onto the top wall and press along its full width; a stuck flap is never peeled back and laid again).
- **Coaching note:** fold it once, press it end to end. If it sticks crooked, leave it — peeling it back kills the strip.

Violation: Accessory left in the returns tote

- **Visible cue:** the episode ends with the coiled cable or the manual card still in the returns tote, or lying loose on the table, on a tray rim, or in the stock bin.
- **SOP rule broken:** Step 5 (both accessories come out of the tote and into their wells, leaving the tote empty).

- **Coaching note:** look in the tote before you restock. It ends the episode empty.

Violation: Accessory in the wrong well, or not seated

- **Visible cue:** the cable is in the card well or the card is in the cable well, an accessory rides a well rim or lies outside the tray, the cable's tie faces down, or the card lies printed side down or crooked to its well.
- **SOP rule broken:** Step 5 (the cable goes in the back cable well tie facing up, the card goes flat in the front card well printed side up and square to it, neither riding a rim).
- **Coaching note:** cable in the back well, card in the front well. Tie up, printing up, flat in the well.

Violation: Accessory dropped in or carried high

- **Visible cue:** the right gripper opens above the accessory tray and the cable or card falls in, it carries an accessory high above the table or the tote rim, or it tips the returns tote to get an accessory out.
- **SOP rule broken:** Step 5 (move one accessory at a time on a low carry, and lower it into its well; never drop it in from above and never tip the tote out).
- **Coaching note:** lift straight up, carry low, lower it in. Nothing gets dropped or poured.

Violation: Unit not restocked, or restocked wrong

- **Visible cue:** the episode ends with the sealed bag still on the bag spot, on the table beside the bin, leaning on the outside of the bin, or riding the bin rim; or it lies in the bin label face down, crooked to the bin walls, or turned end for end from how it lay on the bag spot.
- **SOP rule broken:** Step 6 (lower the sealed bag into the stock bin until it rests on the unit already there, label facing up, square to the bin walls, the same way round as it lay on the bag spot).
- **Coaching note:** carry it level, do not turn your wrist, and take it all the way down before you open the gripper.

Violation: Stock bin not held, or moved off its spot

- **Visible cue:** the left gripper is off the bin rim while the right gripper lowers the bag in, or either gripper drags, lifts, tips, or knocks the stock bin off the front-center.
- **SOP rule broken:** Step 6 (the left gripper presses the bin down by its left rim and stays there until the right gripper is clear).
- **Coaching note:** hold the bin before the bag comes over it, and let go only after the gripper is out.

Violation: Fixture knocked out of place

- **Visible cue:** either gripper tips, drags, or pushes the returns tote, the cloth tray, the label pad, the bag stack, the scrap bin, or the accessory tray off its spot, spills a tray, or lifts the

liner off the label pad.

- **SOP rule broken:** Steps 1–6 (the tote, cloth tray, label pad, bag stack, scrap bin, and accessory tray stay on their spots for the whole episode; grippers reach in over the rim).
- **Coaching note:** reach in over the rim, never move the fixture. Work slower and lower so nothing gets shoved.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence — the right gripper goes to the label pad before the wipe is finished, a bag is laid out before the label is pressed, the unit is bagged before it is wiped, an accessory is moved before the flap is sealed, the bag is restocked before both accessories are in their wells, or the right gripper goes back to wipe or label after the bag is sealed.
- **SOP rule broken:** Steps 1–6 (unbag, wipe, label, bag, reset the accessories, restock, in that order).
- **Coaching note:** finish the step you are in before you start the next one. Unbag, wipe, label, bag, reset, restock.

Violation: Something dropped and left

- **Visible cue:** either gripper drops the unit, a bag, a label, the cloth, the cable, or the card onto the table, into the wrong bin, or onto the floor, and the episode moves on without it being picked up.
- **SOP rule broken:** Steps 1–6 (anything that ends up loose on the table is picked up and put where the step says it belongs before the step goes on).
- **Coaching note:** pick up what you drop before the next move. Nothing stays loose on the table.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper wiping, handling a stock label, sliding the unit into the bag, sealing the flap, moving an accessory, or lifting the sealed bag into the bin; or the right gripper holding the unit's left end, laying out the new bag, binning the used bag, or holding the stock bin rim.
- **SOP rule broken:** Steps 1–6 (the right gripper wipes, labels, bags, seals, resets, and restocks; the left gripper holds the bag, the unit, the mouth, and the bin).
- **Coaching note:** right gripper does the work, left gripper holds. It does not swap.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the restocked bag, the covered label, the sealed flap, the empty tote, and both reset accessories are confirmed; an arm is not home; a gripper is closed; or an arm does something else after homing.

- **SOP rule broken:** Step 7 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- The returns tote arrives without the unit, the cable, or the manual card in it.
- The unit arrives with no old label in its label panel, or with a cracked case.
- The used poly bag arrives still sealed, so the unit cannot be slid out.
- The wipe cloth arrives dry, missing, or torn.
- A stock label will not come off the liner, or has no adhesive left, so it cannot be seated.
- A stock label is misprinted or too small to cover the old label.
- The bag stack is empty, or a new bag's adhesive strip has no tack, so the flap will not seal.
- The stock bin arrives full, so a third bagged unit will not lie flat in it.

Annotation subtasks (from SOP)

1. Lift the bagged unit out of the returns tote and lay it on the bag spot
2. Hold the used bag flat with the left gripper and slide the unit out onto the work spot
3. Drop the used poly bag in the scrap bin with the left gripper
4. Hold the unit at its left end with the left gripper
5. Take the wipe cloth from the cloth tray with the right gripper
6. Make three passes across the top face of the unit
7. Make one pass down the front face of the unit
8. Lay the wipe cloth back flat in the cloth tray
9. Take one stock label off the label pad with the right gripper
10. Lay the stock label over the old label and press it for 2 seconds
11. Lay a new poly bag flat on the bag spot with the left gripper
12. Hold the bag mouth open with the left gripper
13. Slide the unit into the bag with the right gripper
14. Fold the bag flap over and press the seal
15. Hand an accessory from the left gripper to the right gripper above the work spot (Config L and M)

16. Lift the coiled cable from the tote into the cable well
17. Lift the manual card from the tote into the card well
18. Hold the stock bin rim with the left gripper
19. Lift the sealed bag into the stock bin with the right gripper